Earth-like Exoplanets Relational Database

Stage 1: Find and critique a dataset

Intro

When it came to choosing what kind of data was wanted, it was already known that it would be related to astronomy. Astronomy data was chosen before for previous projects visualizing and interpreting galaxy data. This time, planet research seemed like the perfect fit for a relational database because of how we've organized the universe. Planets are part of solar systems, and they are part of galaxies, and they are part of clusters, and they are part of the early universe or the late universe. With some searching, the public NASA exoplanet archive operated by Caltech, under contract with NASA under the Exoplanet Exploration Program, was found (high discoverability), and it felt perfect for the focus of this project, to look for Earth-like planets, not just habitable planets, but ones similar to Earth in as many data points as possible. After getting to the main page for the exoplanet archive, planetary systems and composite data were at the bottom, and in there was a large table with hundreds of columns and rows about planets, easily discoverable and easy to acquire for personal projects through the "Download Table" tab, top left. The data was seen as free and open to the public. Since the focus was on Earth-like planets, constraints were used one by one to see how many planets we could work with. After setting three constraints, radius, equilibrium temperature, and stellar effective temperature, what started as thousands of planets turned into three planets, and all with missing data. If the constraints were too similar to Earth, we would be left with zero data. It was realized that if a database project about Earth-like planets was going to be done, a different approach was needed. So instead, we left out the constraints and downloaded the data as is. We started with two datasets. One for the abbreviations of the columns and the other for the exoplanet data. We started with 320 columns and 6158 planets. The rest, like the cleaning and preprocessing, would be left with the Jupyter notebook. If we start with as much data as possible and then clean up the data in Jupyter, then we'll have more to work with. Additionally, we can choose to balance between complete data and earth-like parameters or find a happy medium where we have complete data and earth-like planets with multiple constraints.

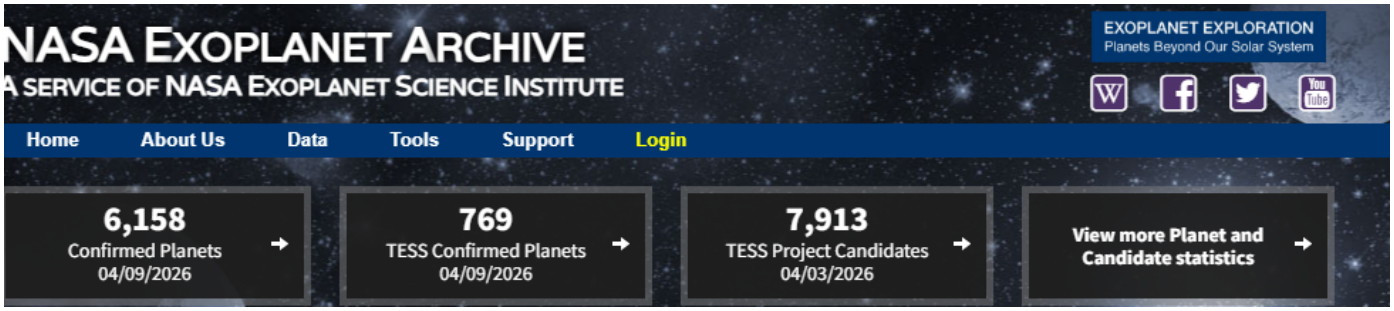


Figure 1. Top of the main page for the exoplanet archive

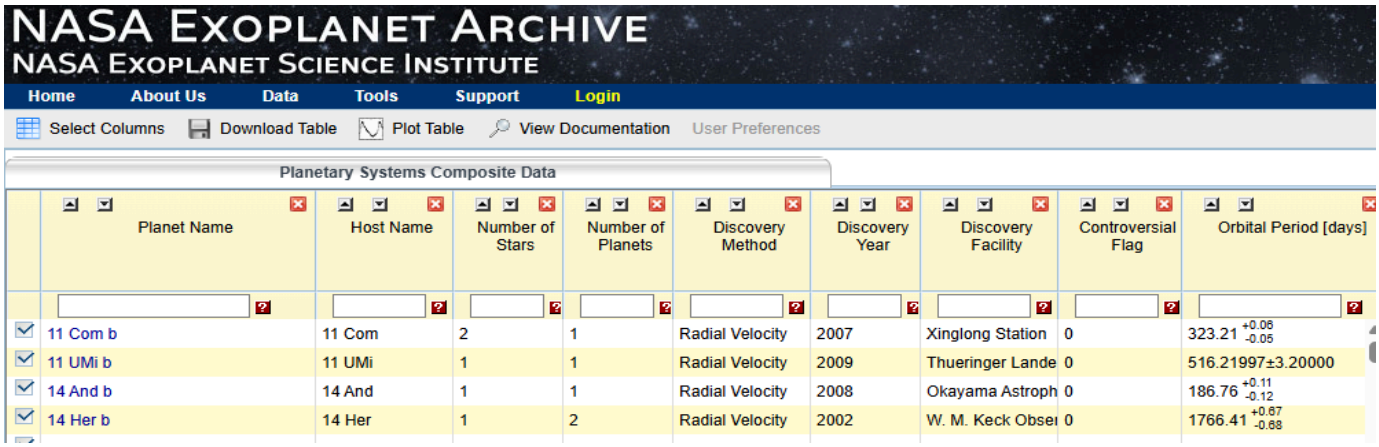


Figure 2. Full exoplanet database

	A	B	C	D	E	F	G	H	I
1	planet_id	pl_name	hostname	pl_letter	hd_name	hip_name	tic_id	gaia_dr2_id	gaia_dr3_ic
2	1	11 Com b	11 Com	b	HD 107383	HIP 60202	TIC 72437047	Gaia DR2 39469	Gaia DR3 3
3	2	11 UMi b	11 UMi	b	HD 136726	HIP 74793	TIC 230061010	Gaia DR2 16967	Gaia DR3 1
4	3	14 And b	14 And	b	HD 221345	HIP 116076	TIC 333225860	Gaia DR2 19201	Gaia DR3 1
5	4	14 Her b	14 Her	b	HD 145675	HIP 79248	TIC 219483057	Gaia DR2 13852	Gaia DR3 1
6	5	16 Cyg B b	16 Cyg B	b	HD 186427	HIP 96901	TIC 27533327	Gaia DR2 21355	Gaia DR3 2
7	6	17 Sco b	17 Sco	b		HIP 79540	TIC 135596590	Gaia DR2 43424	Gaia DR3 4
8	7	18 Del b	18 Del	b	HD 199665	HIP 103527	TIC 354489950	Gaia DR2 17567	Gaia DR3 1

Figure 3. Downloaded the data and opened it into a Google Sheets file, and added the primary key, planet_id

There are 5 questions I want to research, looking at this data.

1. Other than Earth-like size, like a similar radius, what other parameters or constraints should I include that are also included in the column data table, to find an Earth-like planet? Why are those columns significant when looking for Earth-like planets?
2. How many planets from our data fulfill all these parameters to find an Earth-like planet? Does it match what was expected? Why or why not?
3. What would a visit to the planet be like according to the data?
4. Most, if not all, of the planets that have been discovered reside in our own galaxy, the Milky Way. According to the data we have, estimate how many Earth-like planets there are in the Universe based on the number of galaxies that have been discovered? What does this number say about us?
5. What are the biases and limitations of the data, and these questions? And what could it mean for our final estimation?

From there I started preprocessing and cleaning the Earth-like planet data in Jupyter Lab. The data is open, real, and not too simple. There are way too many columns, 320, including many that aren't helpful for this project. We need to find the critical columns that we need and dispose of the rest, but in order to do that, we need to answer research question 1 listed above. "Other than Earth-like size, like a similar radius, what other parameters or constraints should I include that are also included in the column data table, to find an Earth-like planet? Why are those columns significant when looking for Earth-like planets?"

Name box (Ctrl + J)	B	C
1	column_id	abbrev_name	full_name
2	1	planet_id	Planet ID
3	2	pl_name	Planet Name
4	3	hostname	Host Name
5	4	pl_letter	Planet Letter
6	5	hd_name	HD ID
7	6	hip_name	HIP ID
8	7	tic_id	TIC ID
9	8	gaia_dr2_id	Gaia DR2 ID
10	9	gaia_dr3_id	Gaia DR3 ID
11	10	sy_snum	Number of Stars
12	11	sy_pnum	Number of Planets
13	12	sy_mnum	Number of Moons
14	13	cb_flag	Circumbinary Flag
15	14	discoverymethod	Discovery Method
16	15	disc_year	Discovery Year
17	16	disc_refname	Discovery Reference
18	17	disc_pubdate	Discovery Publication Date
19	18	disc_locale	Discovery Locale
20	19	disc_facility	Discovery Facility
21	20	disc_telescope	Discovery Telescope

Figure 4. The column data sheet includes full names for each column. I highlighted the ones most critical

We went through the data sheet and manually highlighted the ones that seemed the most critical for our project. Then, in a Jupyter notebook, we used the highlighted columns and filtered out the data to include only the critical columns in our column data set (20) and in our planets dataset (31).

```
[20]: #### Find critical columns in column data
critical_column_data=['planet_id','pl_name', 'hostname', 'pl_letter', 'sy_snum', 'sy_pnum',
                      'sy_mnum','discoverymethod', 'disc_year', 'disc_facility',
                      'pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bmasse',
                     'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl', 'pl_tranmid',
                     'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_rad', 'st_mass',
                     'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsin', 'st_rotp',
                     'st_radv', 'ra', 'dec', 'sy_dist']

new_column_data = column_data[column_data['abbrev_name'].isin(critical_column_data)]

#new_column_data = column_data[critical_column_data]
print(new_column_data)
```

```
     column_id      abbrev_name                          full_name
0            1        planet_id                          Planet ID
1            2          pl_name                        Planet Name
2            3         hostname                          Host Name
3            4        pl_letter                       Planet Letter
9           10          sy_snum                     Number of Stars
```

Figure 5. Critical columns listed and used to redefine the column dataset

```
[31]: #### Find critical columns
critical_planet_columns=['planet_id','pl_name', 'hostname', 'pl_letter', 'sy_snum', 'sy_pnum',
                         'sy_mnum','discoverymethod', 'disc_year', 'disc_facility',
                         'pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bmasse',
                        'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl', 'pl_tranmid',
                        'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_rad', 'st_mass',
                        'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsin', 'st_rotp',
                        'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data = planet_data[critical_planet_columns]
print(new_planet_data)
```

```
      planet_id     pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0             1    11 Com b    11 Com         b        2        1        0
1             2    11 UMi b    11 UMi         b        1        1        0
2             3    14 And b    14 And         b        1        1        0
3             4    14 Her b    14 Her         b        1        2        0
4             5  16 Cyg B b  16 Cyg B         b        3        1        0
...         ...         ...       ...       ...      ...      ...      ...
```

Figure 6. Used the critical columns to redefine the planets dataset

After cleaning the columns, we looked at the rows or entries. There are hundreds of rows with missing data. We need to clean the data so that only the necessary columns with entries for every row show. To do this, we found out the sum of missing data from each column shown on the left. Then we used those columns to drop all the rows or planets with missing data, so when we look for sums of missing data from each column, we are left with zero for each column.

```
planet_id             0
pl_name               0
hostname              0
pl_letter             0
sy_snum               0
sy_pnum               0
sy_mnum               0
discoverymethod       0
disc_year             1
disc_facility         0
pl_orbper           337
pl_orbsmax          315
pl_angsep           344
pl_rade              50
pl_bmasse            31
pl_dens             139
pl_orbeccen         938
pl_insol           1842
pl_eqt             1565
pl_orbincl         1522
pl_tranmid         1266
pl_trandep         1777
pl_trandur         1683
pl_rvamp           3667
st_spectype        3847
st_rad              317
st_mass               8
st_met              553
st_lum              311
st_logg             321
st_age             1314
st_dens             586
st_vsin            3881
st_rotp            5268
st_radv            3772
ra                    0
dec                   0
sy_dist              27
```

Figure 8. Counted columns with missing values

```
[45]: #calculating rows with missing values again after dropping planets missing data
#all are zeros
missing_column_data = new_planet_data.isna()
total_columns = missing_column_data.sum(axis=1)
#left with 162 entries, or 162 planets that have data in all the required columns
print(total_columns)
```

```
36      0
54      0
55      0
110     0
209     0
       ..
6078    0
6097    0
6100    0
6107    0
6141    0
Length: 162, dtype: int64
```

Figure 7. Checked for missing values in the main dataset, new_planet_data

```
[33]: #we need to drop all planets with missing values under a column, starting with pl_orbper
required_columns = ['pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bmasse',
                    'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl', 'pl_tranmid',
                    'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_rad', 'st_mass',
                    'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsin', 'st_rotp',
                    'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data.dropna(subset= required_columns, inplace=True)

#calculating columns with missing values again after dropping planets missing data
#all are zeros
missing_row_data=new_planet_data.isna()
total_rows=missing_row_data.sum()
print(total_rows)
```

Figure 9. Dropped planets with missing values

When we looked at the number of rows or planets left, we found that we had 162 planets with complete data, as shown in Figure 7. This is a good number to start with for the Earth-like constraints we will use later on.

The other dataset, called planets_column_data, doesn't contain missing values because of its nature. Every column has a full name column. In other words, this dataset is a list of abbreviations and their full name and is used to understand the abbreviated columns in the main planet dataset. It's a pretty simple table with only 3 columns: ID, abbreviation, and full name, as shown below.

```
[22]: #redoing the planet's column id, 1 to length of column, 1 to 38
new_column_data['column_id']=range(1, len(new_column_data) + 1)
print(new_column_data)
```

	column_id	abbrev_name	full_name
0	1	planet_id	Planet ID
1	2	pl_name	Planet Name
2	3	hostname	Host Name
3	4	pl_letter	Planet Letter
4	5	sy_snum	Number of Stars
5	6	sy_pnum	Number of Planets
6	7	sy_mnum	Number of Moons
7	8	discoverymethod	Discovery Method
8	9	disc_year	Discovery Year
9	10	disc_facility	Discovery Facility

Figure 10. Column data set shown above

Stage 2. Model your data

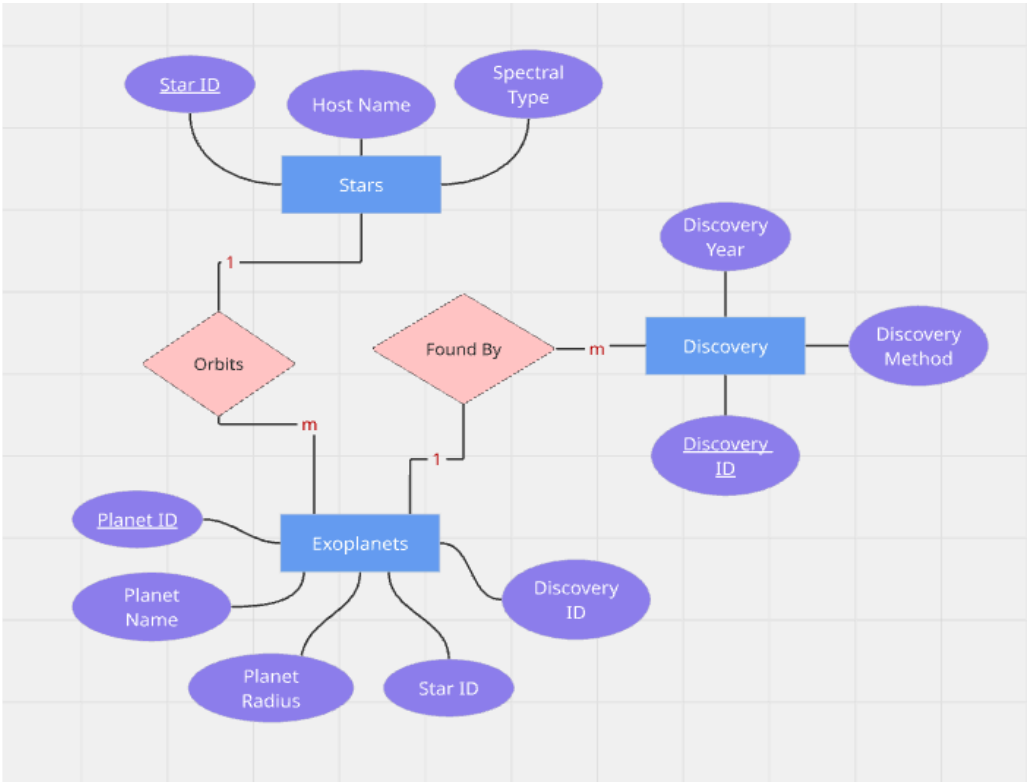


Figure 11. First Basic Entity Relationship Diagram

Entities: Stars, Exoplanets, Discovery
Primary Keys: Planet ID, Star ID, Discovery ID
Exoplanets Attributes: Planet ID, Planet Name, Planet Radius, etc.
Stars Attributes: Star ID, Host Name, Spectral Type, etc.
Discovery Attributes: Discovery ID, Discovery Year, Discovery Method, etc.
Relationships: Exoplanets and Stars- m:1, Exoplanets and Discovery- 1:m

When the model was started, the quick realization that having two datasets was unnecessary and made data handling more complicated than it needed to be was found. It was a lot easier and more direct to just rename the columns in Jupyter Lab and have the full names ready to go in the main dataset for exoplanets. This also made modeling the ER diagram easier. The first basic ER diagram created is shown above. There were a few basic errors caught in the first ER diagram, later on. Firstly, foreign keys discovery ID and star ID don't need to be added as bubbles under other entities. The lines show the relationships between entities and primary keys. Secondly, there can be many exoplanets per discovery, so a many to one relationship should be shown between exoplanets and discovery. Thirdly, Host Name usually refers to the solar system or system name, so it should be under a different entity. Fourthly, a fourth entity called Solar System should be included for system attributes that don't describe the star directly. Finally, all the attributes should be added to the final draft of the ER diagram.

I created a final draft of the ER Diagram shown below that fixes all the problems mentioned and showcases the four main entities in the data: **Exoplanets, Stars, Discoveries, and Solar System**. There are 38 columns of data plus the foreign keys underlined, and each entity has a bubble, making 41 bubbles total. This model could be used to understand the relationships between entities and how the database will be organized. There is a many-to-many relationship between exoplanets and stars, in which multiple exoplanets can orbit multiple stars. There is a many-to-many relationship between exoplanets and discoveries, where multiple exoplanets can happen with many discovery moments, facilities, methods, etc. Finally, there is a many-to-one relationship between stars and solar systems, where a system can have more than one star.

In the ER diagram shown below, we included all the attributes for each entity for a complete diagram and showed all the relationships between entities. Star ID, Discovery ID, and System ID will be used as foreign keys in Exoplanets when creating the database using SQL tables. The new diagram shows how the data will be normalized by splitting the flat data table into separate entities for a relational model database. Attributes don't have to be repeated more than once in multiple entities. Certain bubbles didn't directly depend on stars like distance. Distance is the distance between Earth and the system the planet is part of. If there is an exoplanet that belongs to a certain system, the distance from the Earth to that system won't be repeated twice for stars and the solar system. Systemic Radial Velocity is a bubble under Stars and not the solar system because it uses the star's light to find the velocity at which the system is moving away. Since it depends on the star, it's listed under Stars. When data is organized into separate entities, we avoid the redundancy seen in a flat dataset table. For example, we avoid listing the star data and the solar system data multiple times for multiple exoplanets found in the same system and star. Instead of repeating this data, it is normalized so the data about the star or the solar system only has to be listed once. This makes it easier to grow and manipulate the database as it grows.

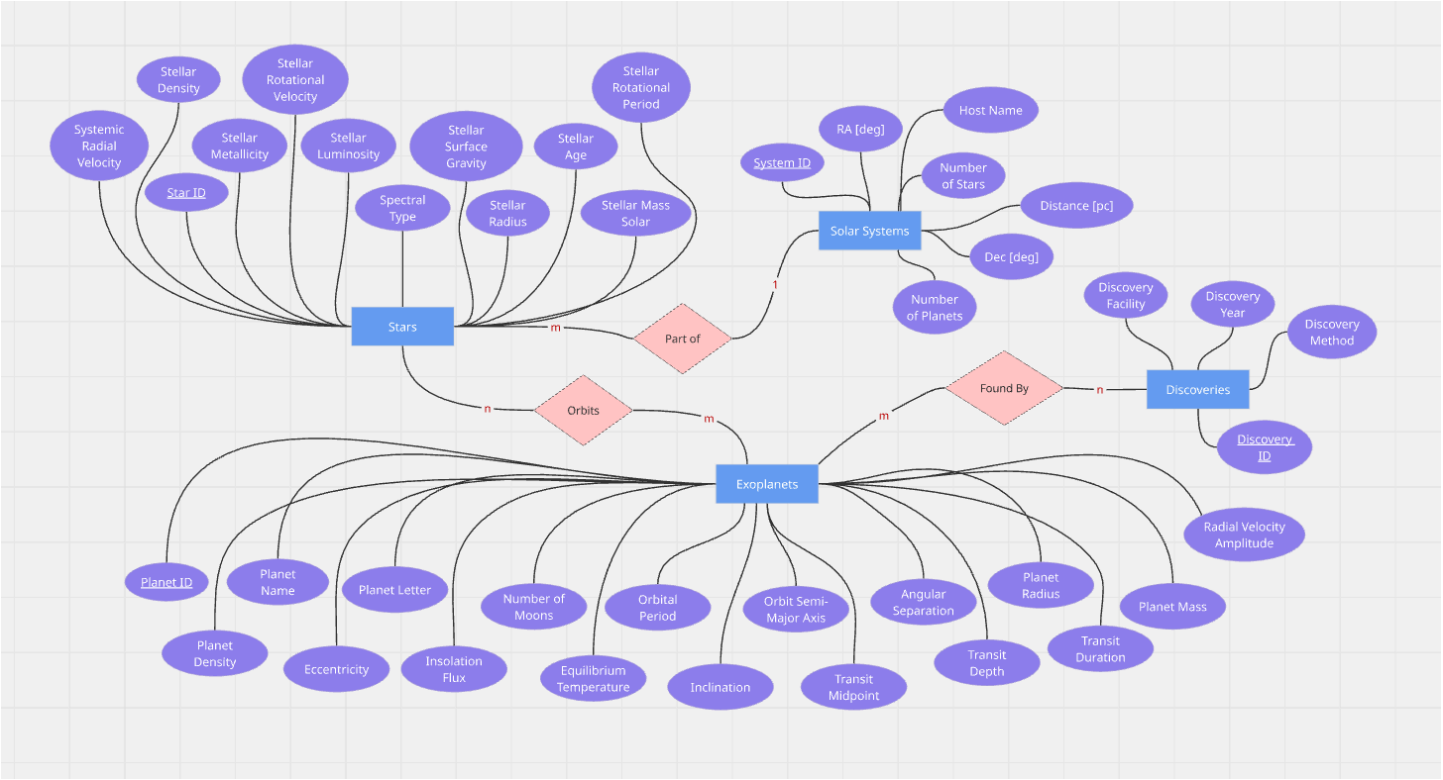


Figure 12. Final ER Diagram

The final part of stage two, as shown in figures 13 to 20, is to add the foreign keys in Jupyter Lab and to rename all the original column names to their actual Full Names to make the implementation on SQL better. We can do this by using the rename() Python function to rename each column and by adding the foreign keys as columns in the exoplanet dataset. We can create 3 new datasets for each entity and then add the appropriate data to the correct dataset, as shown below.

```
## instead of having two datasets, make the columns names into their full names for more straight forward data handling

#rename columns here
new_planet_data = new_planet_data.rename(columns={'planet_id': 'Planet ID', 'pl_name': 'Planet Name',
                                                  'hostname': 'Host Name', 'pl_letter': 'Planet Letter',
                                                  'sy_snum':'Number of Stars', 'sy_pnum':'Number of Planets',
                                                  'sy_mnum':'Number of Moons','discoverymethod':'Discovery Method',
                                                  'disc_year':'Discovery Year','disc_facility': 'Discovery Facility',
                                                  'pl_orbper':'Orbital Period [days]','pl_orbsmax':'Orbit Semi-Major Axi
```

Figure 13. Renamed columns to their full names as found in the columns data set

```
print(new_planet_data)
```

```
   Planet ID Planet Name Host Name Planet Letter  Number of Stars  \
0          1    55 Cnc e    55 Cnc             e                2
1          2    AU Mic b    AU Mic             b                1
2          3    AU Mic c    AU Mic             c                1
3          4  CoRoT-18 b  CoRoT-18             b                1
```

Figure 14. Rename columns result

[21]:
```
## Add data to new data sets
#i need to find the columns, then i need add the columns to the new dataset

#Stars
#Spectral Type, Stellar Radius [Solar Radius],
#Stellar Mass [Solar mass], Stellar Metallicity [dex],
#Stellar Luminosity [log(Solar)], Stellar Surface Gravity [log10(cm/s**2)],
#Stellar Age [Gyr], Stellar Density [g/cm**3], Stellar Rotational Velocity [km/s]
#Stellar Rotational Period [days], Systemic Radial Velocity [km/s],

#find columns and assign to variable
stars_columns=['Spectral Type', 'Stellar Radius [Solar Radius]',
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/s]',
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]']

#redefine stars_data using new varable with column names
stars_data = new_planet_data[stars_columns]

#set variable number of rows in dataset
number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows) at the end of the data
stars_data.insert(0, 'Star ID', range(1,  number_rows + 1))


print(stars_data)
```

```
     Star ID Spectral Type  Stellar Radius [Solar Radius]  \
0          1          G8 V                         0.9430
```

Figure 15. Added the primary keys to the start of each dataset

```
[162 rows x 18 columns]
```

[26]:
```
#count of entries before drop_duplicates
star_entries_before = len(stars_data)
print( 'stars before: ', star_entries_before)

stars_data.drop_duplicates(subset=['Spectral Type', 'Stellar Radius [Solar Radius]',
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/s]',
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]'], inplace=True)

#after drops
star_entries_after = len(stars_data)
print( 'stars after:', star_entries_after)

#difference
star_entries_dropped = star_entries_before - star_entries_after
print("Duplicates Removed:", star_entries_dropped)
```

```
stars before:  162
stars after: 131
Duplicates Removed: 31
```

Figure 16. Duplicates were removed from the stars and the other datasets

```
missing_values = discoveries_data.isna()

row_total=missing_values.sum()
print(row_total)
print('_____Discoveries_____')
```

```
System ID                                  0
dtype: int64
_____Stars_____
System ID           0
Host Name           0
Number of Stars     0
Number of Planets   0
RA [deg]            0
Dec [deg]           0
Distance [pc]       0
dtype: int64
_____Solar System_____
Discovery ID          0
Discovery Method      0
Discovery Year        0
Discovery Facility    0
dtype: int64
_____Discoveries_____
```

Figure 19. No missing values were found in each dataset

```
[30]: #store cleaned datasets in new CSV files

      #save stars data
      cleaned_stars_data = stars_data.copy()
      cleaned_stars_data.to_csv('cleaned_stars_data.csv', index=False)
      print(cleaned_stars_data)
```

Figure 20. After preprocessing and cleaning, each dataset for each entity was saved into a CSV file for later use in creating the database using SQL

The main exoplanet data was split into four entities or four datasets. For each dataset, we defined the critical columns we wanted from the exoplanets dataset and then redefined the new dataset using the critical columns we listed, so all the columns listed would be taken from the original dataset and copied over to the new dataset. After taking the critical columns for stars, systems, and discoveries, we were left with exoplanets, so we could redefine exoplanets in the end without missing columns.

We used drop_duplicates() to remove duplicate entries for the same star, system, discovery, or planet. We used isna() to find missing values. We added foreign keys System ID, Star ID, and Discovery ID to the exoplanet dataset, and we added a foreign key, System ID, to the Stars dataset to link it to the solar system it's in.

After all the cleaning, processing, and splitting data into datasets, we normalized our data and achieved 3NF, 3rd Normal Form. Then all the datasets were saved into new CSV files for later use in the next stage of the database creation. All the tables will include the full names of each attribute and the primary keys and foreign keys.

```
#merging or linking tables Stars with Systems before MySQL Lab, one to Many
# we can use map() and dict() because system_data is already normalized
#and each row has a unique system or host name

#AUTOMATICALLY build the mapping dictionary from your stars DataFrame

#This pairs every star name with its MySQL-ready ID, no matter how many hundreds there are

system_id_lookup = dict(zip(system_data['Host Name'], system_data['System ID']))

#Use .map() to fill the entire 'star_id' column at once

stars_data['System ID'] = stars_data['Host Name'].map(system_id_lookup)

#Clean up: Drop the text star_name column so MySQL stays perfectly relational

print(stars_data)
```

```
     Star ID Host Name  System ID Spectral Type  \
0          1    55 Cnc          1          G8 V
1          2    AU Mic          2          M1 V
3          3  CoRoT-18          3          G9 V
4          4    G 9-40          4        M2.5 V
5          5   Gl 1132          5        M4.5 V
```

Figure 21. Setting up the relationships and tables for MySQL, One-to-Many, One solar system to many stars.

```
[38]: #merging or linking tables Stars and Exoplanets before MySQL Lab
      # creating junction data sets for junction data tables, Many to Many

      junction = pd.merge(
          new_planet_data,
          stars_data,
          on = 'Host Name',
          how= 'inner'
      )

      Star_Exoplanet_Orbits = junction[['Star ID','Planet ID']]

      print(Star_Exoplanet_Orbits)
```

```
     Star ID  Planet ID
0          1          1
1          2          2
2          2          3
3          3          4
4          4          5
..       ...        ...
160      127        158
161      128        159
162      129        160
163      130        161
164      131        162
```

Figure 22. Setting up junction tables for MySQL, Exoplanets, and Stars, Many to Many

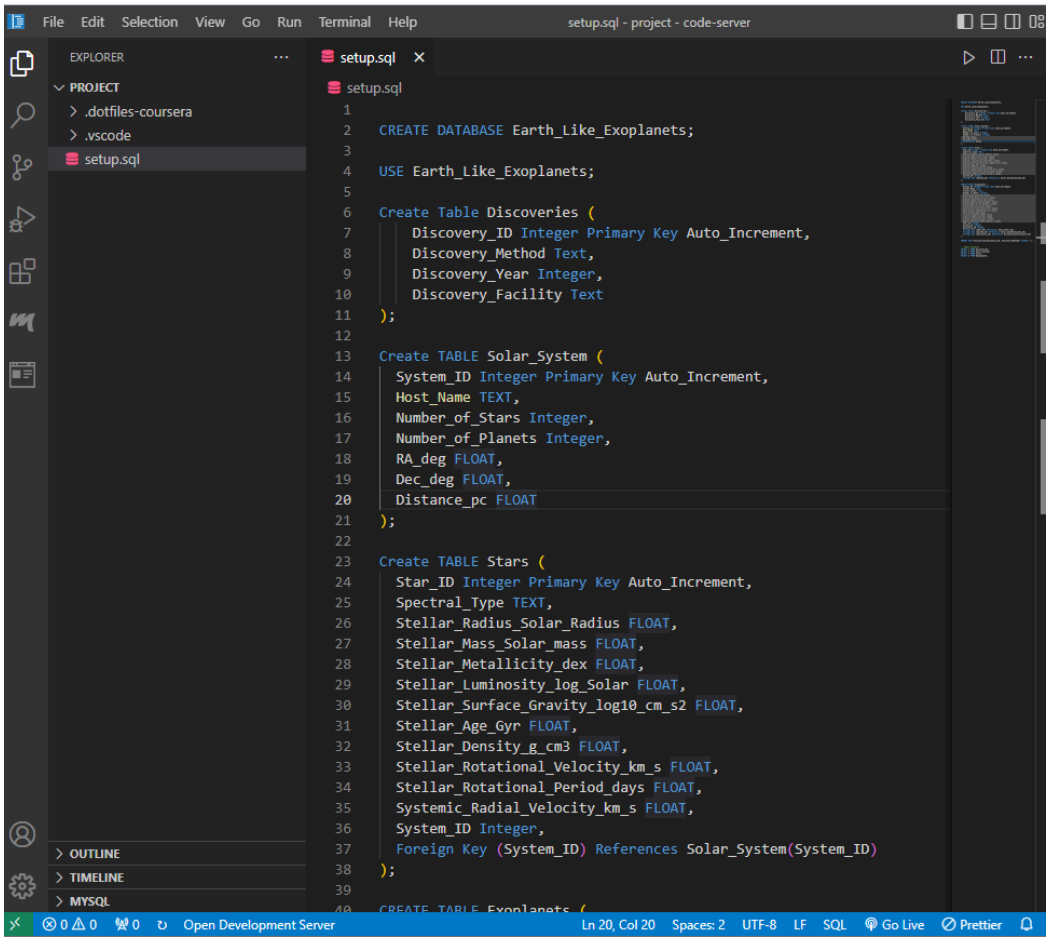
After we had normalized and cleaned the data, we were left with four datasets but no way of linking them with foreign keys and primary keys. After much trial and error, we found that we needed two junction tables where two datasets can merge to prevent duplication or keep the normalization. It was found that junction tables are typically used for the many-to-many relationships. If we can imagine a Venn diagram where exoplanets have matching stars that they orbit, then we can understand that the middle part of the Venn diagram, or inner join, is where two datasets are linked or share data. The junction table in Figure 22 is for stars and exoplanets, and

the other junction table is for exoplanets and discoveries. At the end, we created and saved 6 different datasets: exoplanets, stars, solar systems, discoveries, stars & exoplanets, and exoplanet & discoveries. These are ready to use for MySQL for creating our database.

Stage 3. Create the database

```sql
CREATE TABLE Solar_System (
  System_ID INTEGER PRIMARY KEY AUTOINCREMENT,
  System_Name TEXT,
  Number_of_Planets INTEGER,
  Star_ID INTEGER,
  FOREIGN KEY (Star_ID) REFERENCES Stars(Star_ID)
);
CREATE TABLE Stars (
  Star_ID INTEGER PRIMARY KEY AUTOINCREMENT,
  Host_Name TEXT,
  Spectral_Type TEXT
);
CREATE TABLE Exoplanets (
  Planet_ID INTEGER PRIMARY KEY AUTOINCREMENT,
  Planet_Name TEXT,
  Radius INTEGER,
  Star_ID INTEGER,
  System_ID INTEGER,
  FOREIGN KEY (Star_ID) REFERENCES Stars(Star_ID),
  FOREIGN KEY (System_ID) REFERENCES Solar_System(System_ID));
```

Figure 23. First SQLite Rough Model created in SQL Fiddle as a preliminary rough draft to practice SQL




```sql
CREATE DATABASE Earth_Like_Exoplanets;

USE Earth_Like_Exoplanets;

Create Table Discoveries (
    Discovery_ID Integer Primary Key Auto_Increment,
    Discovery_Method Text,
    Discovery_Year Integer,
    Discovery_Facility Text
);

Create TABLE Solar_System (
  System_ID Integer Primary Key Auto_Increment,
  Host_Name TEXT,
  Number_of_Stars Integer,
  Number_of_Planets Integer,
  RA_deg FLOAT,
  Dec_deg FLOAT,
  Distance_pc FLOAT
);

Create TABLE Stars (
  Star_ID Integer Primary Key Auto_Increment,
  Spectral_Type TEXT,
  Stellar_Radius_Solar_Radius FLOAT,
  Stellar_Mass_Solar_mass FLOAT,
  Stellar_Metallicity_dex FLOAT,
  Stellar_Luminosity_log_Solar FLOAT,
  Stellar_Surface_Gravity_log10_cm_s2 FLOAT,
  Stellar_Age_Gyr FLOAT,
  Stellar_Density_g_cm3 FLOAT,
  Stellar_Rotational_Velocity_km_s FLOAT,
  Stellar_Rotational_Period_days FLOAT,
  Systemic_Radial_Velocity_km_s FLOAT,
  System_ID Integer,
  Foreign Key (System_ID) References Solar_System(System_ID)
);
```

Figure 24. Started writing Create commands for each table and junction

We created four main tables for the four main entities. We followed the W3School's SQL tutorial for syntax and conventions along with class lectures. We ran into trouble with the naming of the columns. Turns out, if we wanted to create a column with the units typically found in physics to describe the quantities, we could not. Instead, we had to use underscores and take away all symbols that are not allowed in SQL syntax, so some units that are shown in the schema seem incorrect.

We added DROP DATABASE at the top, which drops the database when the script is rerun. Create Database was next, with the name listed, Earth_Like_Exoplanets. Then, USE Earth_Like_Exoplanets is used so the system knows which context to use for commands, so it doesn't mistake the database or table. Then we used Create table for each table or entity. The column names were then listed beneath the table name in the same order they are in the data file, and the data type of each column was specified beside it. After creating every table, including junction tables, it looked like the one below before we ran into some issues. There are several mistakes in the schema below.

—----------

```
DROP DATABASE IF EXISTS Earth_Like_Exoplanets;

CREATE DATABASE Earth_Like_Exoplanets;

USE Earth_Like_Exoplanets;

Create Table Discoveries (
    Discovery_ID Integer Primary Key Auto_Increment,
    Discovery_Method Text,
    Discovery_Year Integer,
    Discovery_Facility Text
);

Create TABLE Solar_System (
  System_ID Integer Primary Key Auto_Increment,
  Host_Name TEXT,
  Number_of_Stars Integer,
  Number_of_Planets Integer,
  RA_deg FLOAT,
  Dec_deg FLOAT,
  Distance_pc FLOAT
);

Create TABLE Stars (
  Star_ID Integer Primary Key Auto_Increment,
  Spectral_Type TEXT,
  Stellar_Radius_Solar_Radius FLOAT,
  Stellar_Mass_Solar_mass FLOAT,
  Stellar_Metallicity_dex FLOAT,
  Stellar_Luminosity_log_Solar FLOAT,
  Stellar_Surface_Gravity_log10_cm_s2 FLOAT,
  Stellar_Age_Gyr FLOAT,
  Stellar_Density_g_cm3 FLOAT,
  Stellar_Rotational_Velocity_km_s FLOAT,
  Stellar_Rotational_Period_days FLOAT,
  Systemic_Radial_Velocity_km_s FLOAT,
  System_ID Integer,
  Foreign Key (System_ID) References Solar_System(System_ID)
```

```
);

CREATE TABLE Exoplanets (
  Planet_ID INTEGER Primary Key Auto_Increment,
  Planet_Name Text,
  Planet_Letter Text,
  Number_of_Moons Integer,
  Orbital_Period_days FLOAT,
  Orbit_Semi_Major_Axis_au FLOAT,
  Angular_Separation_mas FLOAT,
  Planet_Radius_Earth_Radius FLOAT,
  Planet_Mass_or_Earth_Mass FLOAT,
  Planet_Density_g_cm3 FLOAT,
  Insolation_Flux_Earth_Flux FLOAT,
  Equilibrium_Temperature_K FLOAT,
  Inclination_deg FLOAT,
  Transit_Midpoint_days FLOAT,
  Transit_Depth_percent FLOAT,
  Transit_Duration_hours FLOAT,
  Radial_Velocity_Amplitude_m_s FLOAT
);

/*junction  tables*/

CREATE TABLE Star_Exoplanet_Orbits (
  Star_ID INTEGER,
  Planet_ID INTEGER,
  PRIMARY KEY (Star_ID, Planet_ID),
  FOREIGN KEY (Star_ID) REFERENCES Stars(Star_ID),
  FOREIGN KEY (Planet_ID) REFERENCES Exoplanets(Planet_ID)
);

CREATE TABLE Exoplanet_Discoveries (
  Discovery_ID INTEGER,
  Planet_ID INTEGER,
  PRIMARY KEY (Discovery_ID, Planet_ID),
  FOREIGN KEY (Discovery_ID) REFERENCES Discoveries(Discovery_ID),
  FOREIGN KEY (Planet_ID) REFERENCES Exoplanets(Planet_ID)
);

-- insert instance data

INSERT INTO Discoveries(Discovery_Method, Discovery_Year, Discovery_Facility)
VALUES ('Radial Velocity', 2004, 'McDonald Observatory');

/* use command below to run setup.sql again, used to run the script again or to show updates:
mysql -u root < setup.sql */
```

—----------

First, I wanted to use an example to add to the database and see if the functionality was there, so I created a data instance and inserted it into the Discoveries Table. After encountering some syntax errors in the terminal, some SQL commands were rewritten. The SELECT command was used to show the Discovery table below with the instance data.

```
Database changed
mysql> SELECT * FROM Discoveries;
+--------------+------------------+----------------+----------------------+
| Discovery_ID | Discovery_Method | Discovery_Year | Discovery_Facility   |
+--------------+------------------+----------------+----------------------+
|            1 | Radial Velocity  |            204 | McDonald Observatory |
+--------------+------------------+----------------+----------------------+
1 row in set (0.00 sec)

mysql> exit
Bye
```

Figure 25. Showing the instance data added before loading data file

setup.sql

```sql
CREATE TABLE Exoplanets (
  Planet_ID INTEGER Primary Key Auto_Increment,
  Planet_Name Text,
  Planet_Letter Text,
  Number_of_Moons Integer,
  Orbital_Period_days FLOAT,
  Orbit_Semi_Major_Axis_au FLOAT,
  Angular_Separation_mas FLOAT,
  Planet_Radius_Earth_Radius FLOAT,
  Planet_Mass_or_Earth_Mass FLOAT,
  Planet_Density_g_cm3 FLOAT,
  Insolation_Flux_Earth_Flux FLOAT,
  Equilibrium_Temperature_K FLOAT,
  Inclination_deg FLOAT,
  Transit_Midpoint_days FLOAT,
  Transit_Depth_percent FLOAT,
  Transit_Duration_hours FLOAT,
  Radial_Velocity_Amplitude_m_s FLOAT
);

--junction tables below

CREATE TABLE Star_Exoplanet_Orbits (
  Star_ID Integer,
  Planet_ID Integer,
  Primary Key (Star_ID, Planet_ID),
  Foreign Key (Star_ID) References Stars(Star_ID),
  Foreign Key (Planet_ID) References Exoplanets(Planet_ID)

);

CREATE TABLE Exoplanet_Discoveries (
  Discovery_ID Integer,
  Planet_ID Integer,
  Primary Key (Discovery_ID, Planet_ID),
  Foreign Key (Discovery_ID) References Discoveries(Discovery_ID),
  Foreign Key (Planet_ID) References Exoplanets(Planet_ID)

);
```

Figure 26. Fixed Syntax Errors and added Junction Tables to link all tables, junction tables represent many to many relationships seen in the ER diagram in Figure 12.

More mistakes I made shown in the schema above

1. Order, schema should be parents first and then child tables, the schema above shows exoplanets table going after the stars table. This order is incorrect because the parent tables should go before the child tables. Basically, parent tables are independent and don't have foreign keys while the child tables have dependencies like foreign keys. Star table should be the last table in the schema before the junction tables.
2. System ID should be the second column or the second row in the Stars table since it is listed in the data file as the second column. The data will be loaded in that order and will fail to load data if it doesn't match the column name and data type.
3. The Discoveries table had a problem with the formatting shown in the terminal whenever it fills in the discovery method data. Could not figure out how to fix the formatting issues. No other small table has the same formatting issues as long as they have enough space in the terminal for every column.
4. After inserting data using LOAD DATA infile, I had trouble finding the data in the terminal. Sometimes when I put something like SELECT * FROM Solar_System, it would say the table had 0 data after I had submitted a delete data command in previous session. I restarted the terminal but didn't know how to make sure that the database gets rewritten for every session to avoid these issues with data missing. I did some research on google and found a command that would rewrite the database when submitted. I just have to submit the following before every session: mysql -u root < setup.sql . The file runs DROPS DATABASE IF EXISTS and then, CREATE DATABASE … so all tables are reloaded with all the data files.

After fixing these mistakes, the updated version is shown below :

```
—----------------------------------------------------------------------------------------
DROP DATABASE IF EXISTS Earth_Like_Exoplanets;

CREATE DATABASE Earth_Like_Exoplanets;

USE Earth_Like_Exoplanets;

Create Table Discoveries (
   Discovery_ID INTEGER PRIMARY KEY,
   Discovery_Facility TEXT,
   Discovery_Year Integer,
   Discovery_Method VARCHAR(100)
);

/* LOAD DATA*/
LOAD DATA INFILE '/home/coder/project/cleaned_discoveries_data.txt'
INTO TABLE Discoveries
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

/* comments start
-- —---------- instance data

INSERT INTO Discoveries(Discovery_Year, Discovery_Facility, Discovery_Method)
```

```
VALUES ('Radial Velocity', 2004, 'McDonald Observatory');
—----------

Commands to try:

SELECT Discovery_Method
FROM Discoveries
WHERE Discovery_ID < 10;


SELECT Discovery_ID, Discovery_Method
FROM Discoveries
WHERE Discovery_ID < 10;

SELECT Discovery_ID, Discovery_Facility
FROM Discoveries
WHERE Discovery_ID < 10;

Comments end
*/

Create TABLE Solar_System (
  System_ID INTEGER PRIMARY KEY,
  Host_Name TEXT,
  Number_of_Stars Integer,
  Number_of_Planets Integer,
  RA_deg FLOAT,
  Dec_deg FLOAT,
  Distance_pc FLOAT
);

LOAD DATA INFILE '/home/coder/project/cleaned_system_data.txt'
INTO TABLE Solar_System
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

/*
SELECT * FROM Solar_System;
*/

CREATE TABLE Exoplanets (
  Planet_ID INTEGER Primary Key Auto_Increment,
  Planet_Name Text,
  Planet_Letter Text,
  Number_of_Moons Integer,
  Orbital_Period_days FLOAT,
  Orbit_Semi_Major_Axis_au FLOAT,
```

```
  Angular_Separation_mas FLOAT,
  Planet_Radius_Earth_Radius FLOAT,
  Planet_Mass_or_Earth_Mass FLOAT,
  Planet_Density_g_cm3 FLOAT,
  Eccentricity FLOAT,
  Insolation_Flux_Earth_Flux FLOAT,
  Equilibrium_Temperature_K FLOAT,
  Inclination_deg FLOAT,
  Transit_Midpoint_days FLOAT,
  Transit_Depth_percent FLOAT,
  Transit_Duration_hours FLOAT,
  Radial_Velocity_Amplitude_m_s FLOAT
);

LOAD DATA INFILE '/home/coder/project/cleaned_planet_data.txt'
INTO TABLE Exoplanets
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

/*
Comments start

SELECT * FROM Exoplanets;

SELECT Planet_ID, Planet_Name, Number_of_Moons, Planet_Mass_or_Earth_Mass
FROM Exoplanets
WHERE Planet_ID < 100;

Comments end
*/

Create TABLE Stars (
  Star_ID Integer Primary Key Auto_Increment,
  System_ID Integer,
  Spectral_Type TEXT,
  Stellar_Radius VARCHAR(50),
  Stellar_Mass_Solar_mass DOUBLE,
  Stellar_Metallicity_dex FLOAT,
  Stellar_Luminosity_log_Solar FLOAT,
  Stellar_Surface_Gravity_log10_cm_s2 FLOAT,
  Stellar_Age_Gyr FLOAT,
  Stellar_Density_g_cm3 FLOAT,
  Stellar_Rotational_Velocity_km_s FLOAT,
  Stellar_Rotational_Period_days FLOAT,
  Systemic_Radial_Velocity_km_s FLOAT,
  Foreign Key (System_ID) References Solar_System(System_ID)
);
```

```
LOAD DATA INFILE '/home/coder/project/cleaned_stars_data.txt'
INTO TABLE Stars
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

/* Comments start

SELECT * FROM Stars;

SELECT Star_ID, System_ID, Spectral_Type, Stellar_Radius
FROM Stars
WHERE Star_ID < 100;

Comments end
*/

/*
junction  tables
*/

CREATE TABLE Star_Exoplanet_Orbits (
  Star_ID INTEGER,
  Planet_ID INTEGER,
  PRIMARY KEY (Star_ID, Planet_ID),
  FOREIGN KEY (Star_ID) REFERENCES Stars(Star_ID),
  FOREIGN KEY (Planet_ID) REFERENCES Exoplanets(Planet_ID)
);

LOAD DATA INFILE '/home/coder/project/Star_Exoplanet_Orbits_cleaned.txt'
INTO TABLE Star_Exoplanet_Orbits
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

/*
SELECT * FROM Star_Exoplanet_Orbits;
/*

CREATE TABLE Exoplanet_Discoveries (
  Discovery_ID INTEGER,
  Planet_ID INTEGER,
  PRIMARY KEY (Discovery_ID, Planet_ID),
  FOREIGN KEY (Discovery_ID) REFERENCES Discoveries(Discovery_ID),
  FOREIGN KEY (Planet_ID) REFERENCES Exoplanets(Planet_ID)
);
```

```
LOAD DATA INFILE '/home/coder/project/Exoplanet_Discoveries_cleaned.txt'
INTO TABLE Exoplanet_Discoveries
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

/*
SELECT * FROM Exoplanet_Discoveries;
*/

/* Comments start

use command below to run setup.sql again:

$ mysql -u root < setup.sql

go to mysql

$ mysql

mysql> USE Earth_Like_Exoplanets;

Comments end
*/
```

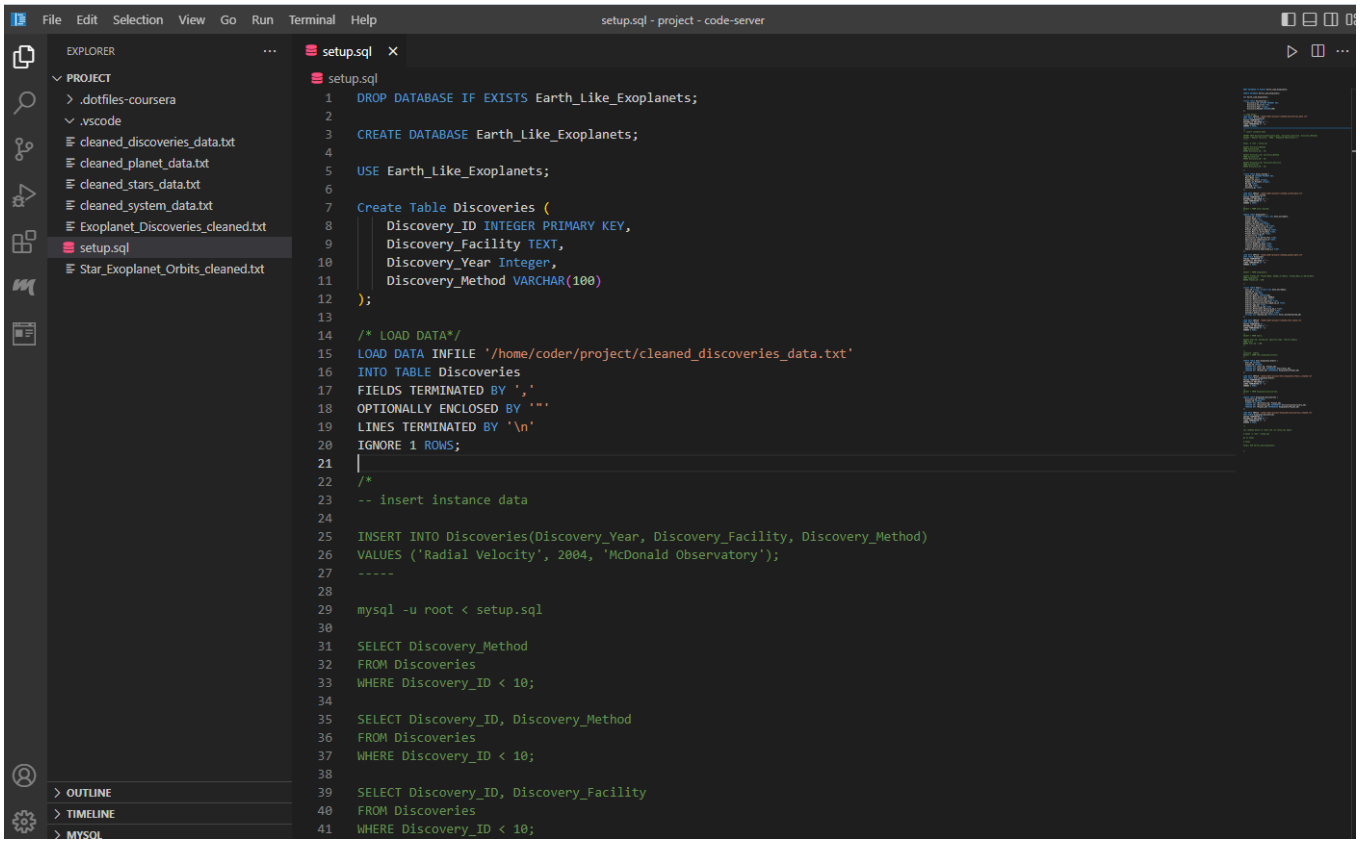


Figure 27. Updated setup.sql with corrections and data files uploaded

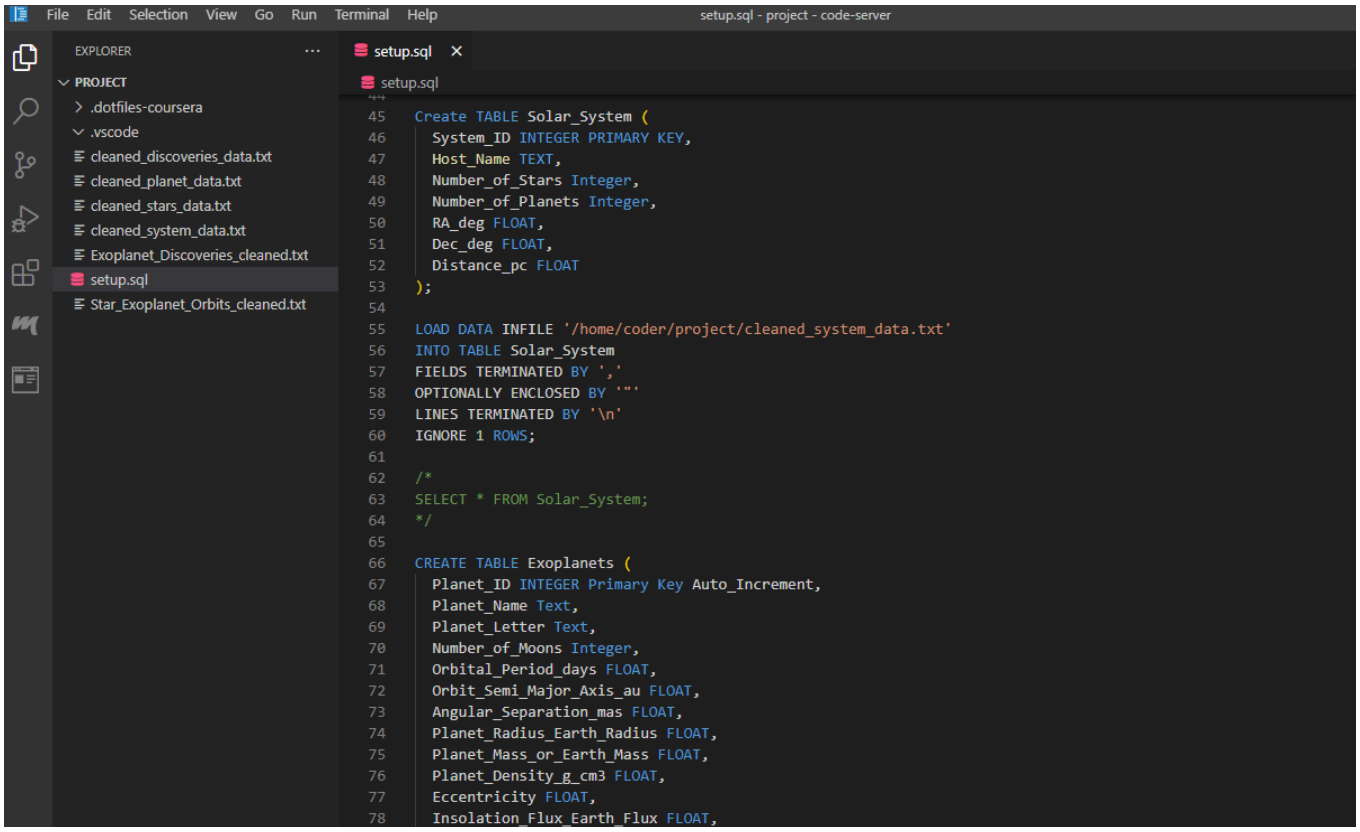


Figure 28. Updated setup.sql with corrections and data files uploaded

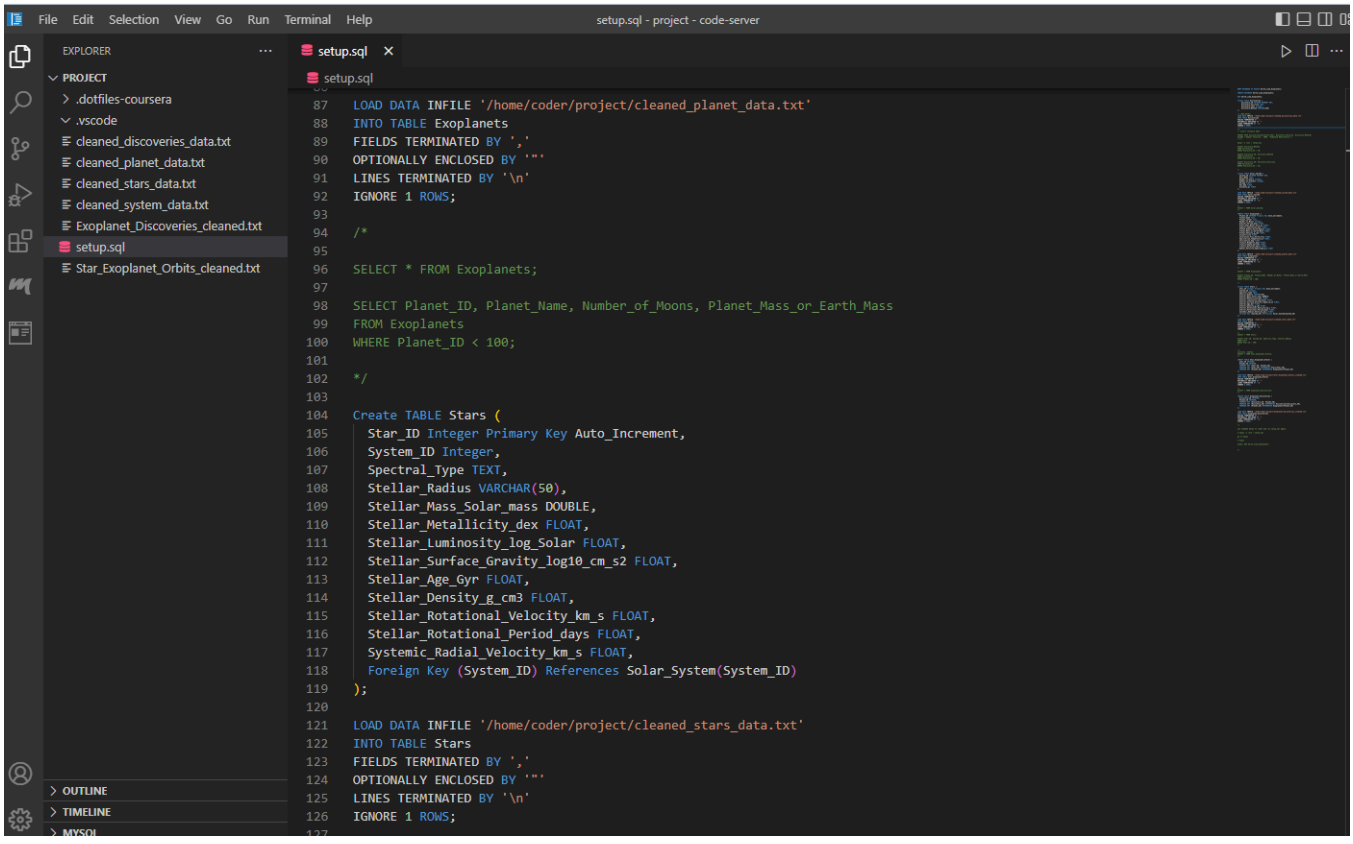


Figure 29. Updated setup.sql with corrections and data files uploaded

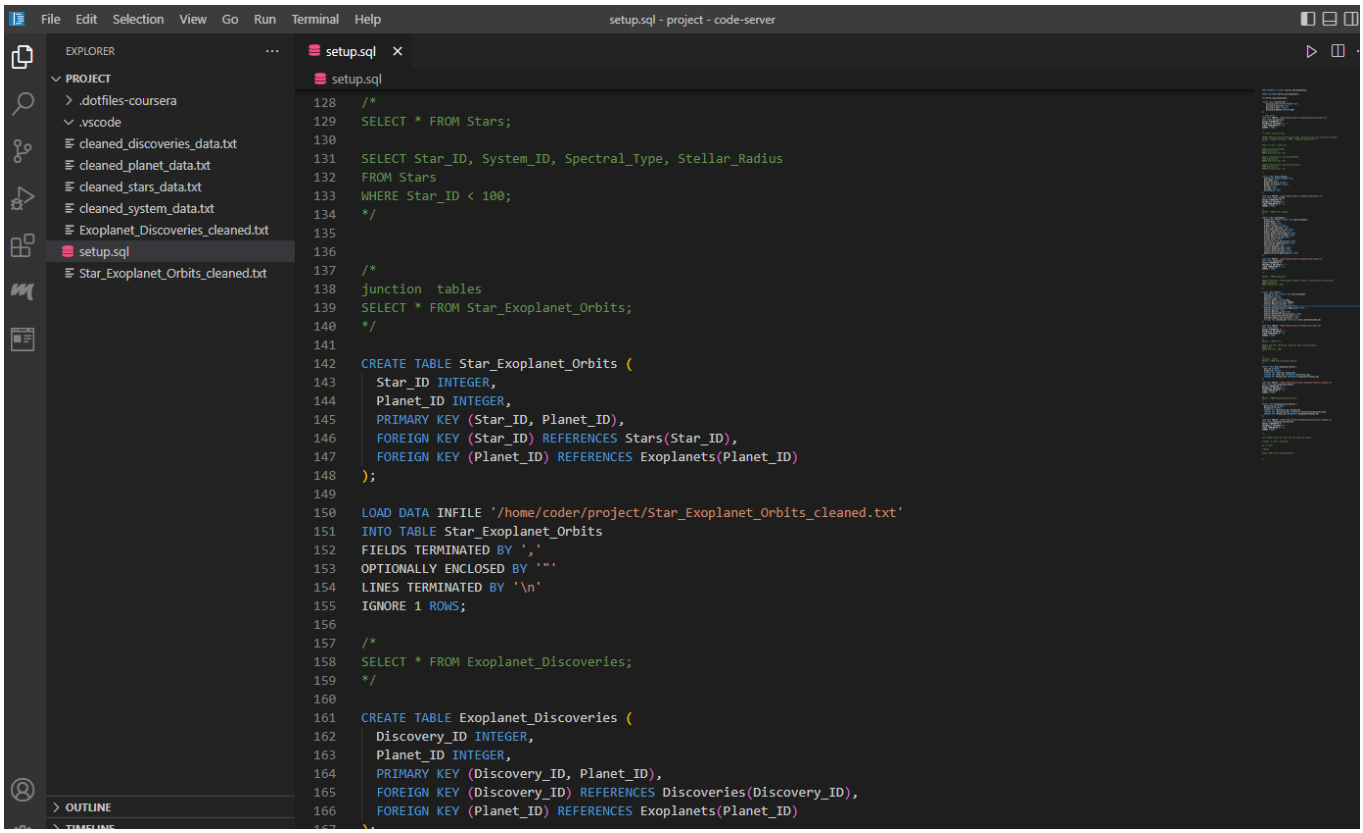


Figure 30. Updated setup.sql with corrections and data files uploaded

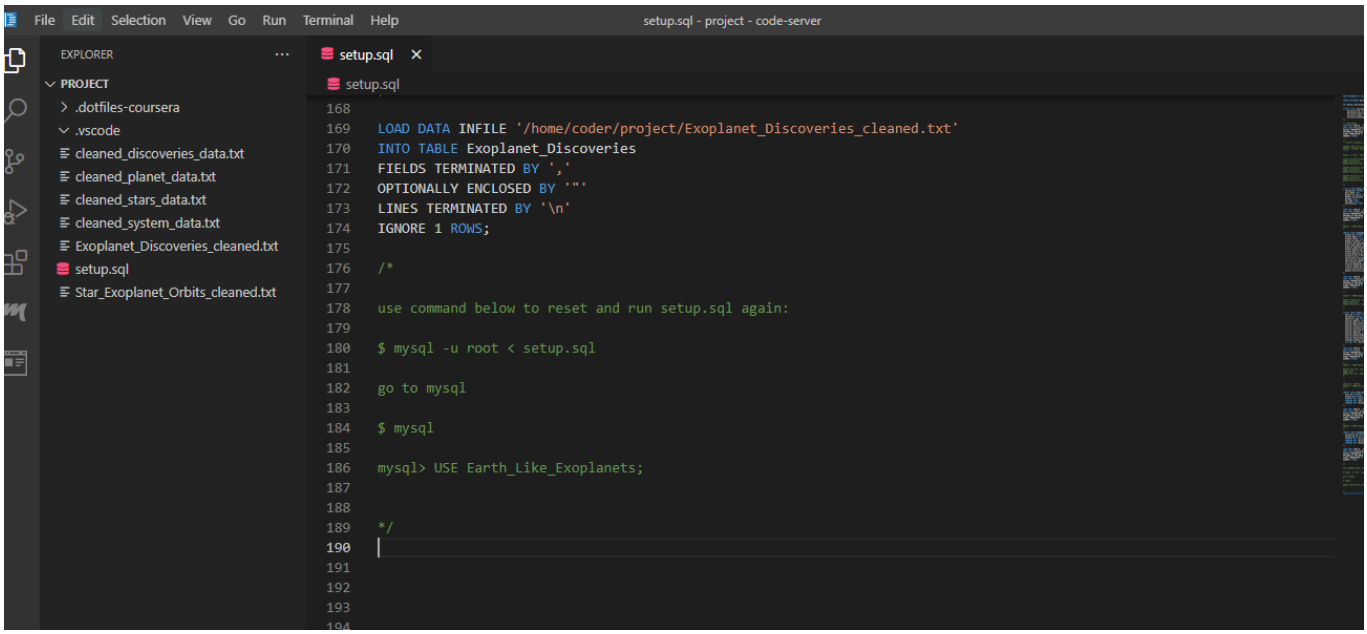


Figure 31. Updated setup.sql with corrections and data files uploaded

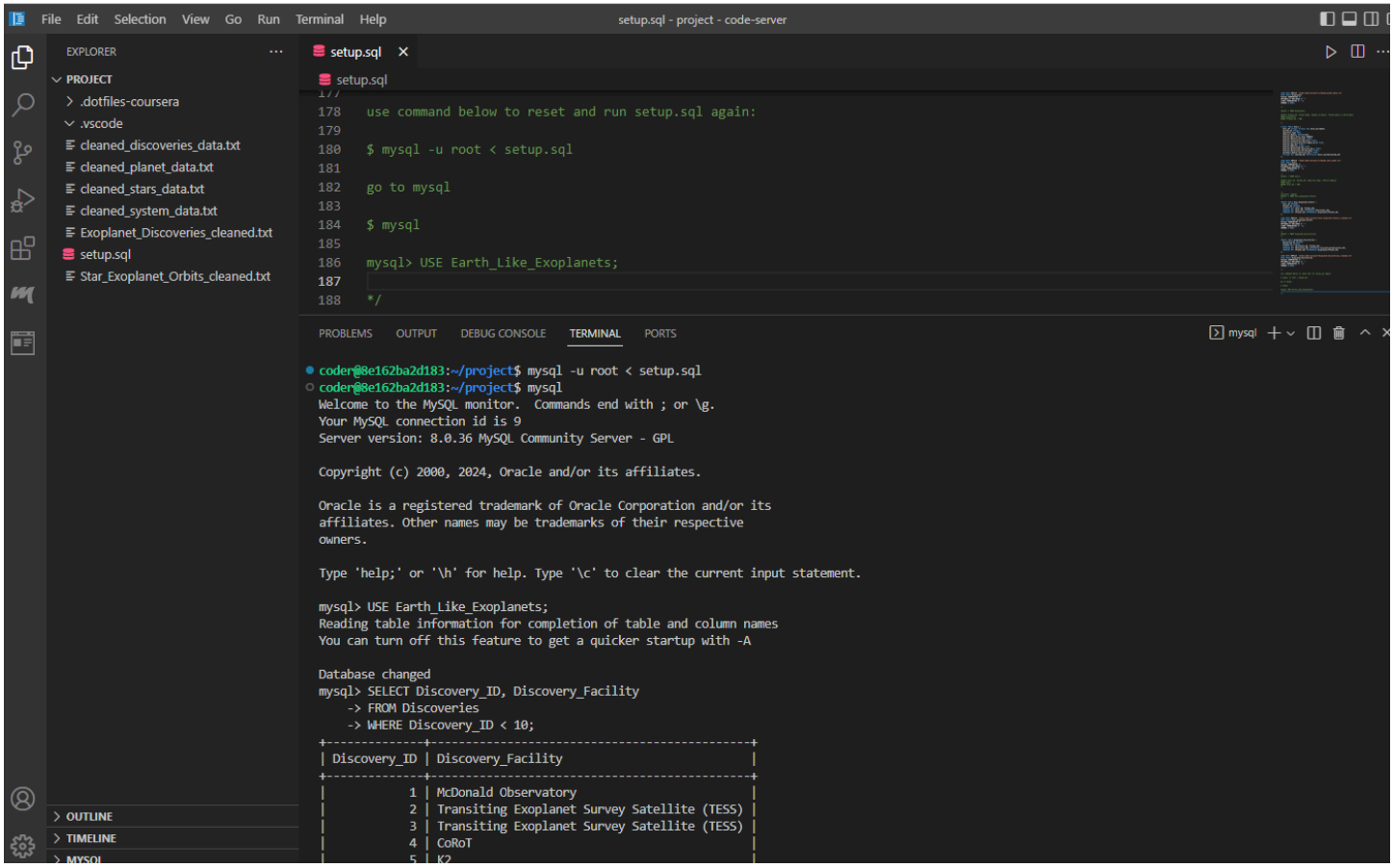


Figure 32. Showing the data that was uploaded in the table Disoveries. All of the tables loaded the data correctly.

Things to point out in this SQL relational database that was done differently than others. All of the data sets or dataframes were normazlied, preprocessed, and cleaned in Jupyter Lab using Python, not in SQL. Some projects normalize and clean the data inside SQL but it was decided that preparing the data as much as possible would make the upload to tables easier considering the initial size of the data. The datasets were also linked with primary keys and foreign keys at the end of the jupyter lab so they would be linked before uploading to the tables. Because the exoplanets data was vast and included. It should be pointed out that we started with 6158 rows or entries in one huge data file and after cleaning it we were left with 162 rows or entries. That is because we deleted all the duplicate entries, all the rows with missing data, and all the entries that weren't distinct. Lastly, for checking all the references in the data, checking every entry manually would take too long. In order to check that all the primary keys and foreign keys properly linked, we submitted the data files to an AI chatbot so it could quickly cross reference everything quickly.

References

https://skyvia.com/blog/how-to-import-csv-file-into-mysql/

https://medium.com/@petehouston/import-csv-file-into-mysql-2f5666205b83